

Question 1:

- 6 Pippa owns a garden construction company. She is preparing a quotation for constructing a garden on a 1000m² site. Her quotation will need to include an estimate of how long the construction will take.

The tasks, together with limitations on their starting times and durations, are as follows:

- A Produce a detailed survey of the site. This must be done before anything else, except that clearing the site can be started at the same time. The surveying will take 1 day.
- B Clear the site. This will take 3 days.
- C Put forward a plan, debate it and adjust it. This can't be done until the survey is complete. It involves meeting with the client, and will take 3 days.
- D Build walls and other fixed features. This cannot be started until the site is cleared and the plan agreed. It will take 3 days.
- E Have a specialist contractor install the pond. This cannot be started until the site is cleared and the plan agreed. It will take 5 days.
- F Plant the trees. This cannot be started until the walls and other fixed features are completed. It will take 3 days.
- G Plant the plants. This cannot be started until the walls and other fixed features are completed. It will take 2 days.
- H Install the irrigation. This cannot be started until the trees are planted and the plants are planted. It will take 2 days.

(i) Draw an activity-on-arc precedence network for this project. [4]

(ii) Complete a forward pass and a backward pass to determine the minimum completion time and the critical activities, using the durations given above. [6]

Activities D and E will be contracted to specialist companies. Pippa plans to complete activities A, B, C, F, G and H herself, but she can employ her friend, Afzal, to do or to help with any of these tasks except for A and C. Each of B, F, G and H can be done by two people, with the total time taken by the two of them being the same as the activity duration.

(iii) The project is to be completed in the shortest time. Produce a schedule to achieve this whilst using as little of Afzal's time as possible. Show who does what and when. Give the minimum completion time and give the total time for which Afzal must be employed. [5]

(iv) How long will it take to complete the project if Pippa does not use Afzal at all? [1]

Question 2:

6

- 5 A village amateur dramatic society is planning its annual pantomime. Three rooms in the village hall have been booked for one evening per week for 12 weeks. The following activities must take place. Their durations are shown.

	Activity	Duration (weeks)
A	choose lead actors	1
B	choose rest of actors	1
C	choose dancers	1
D	rehearse lead actors	8
E	rehearse rest of actors	6
F	rehearse dancers	6
G	prepare scenery	6
H	install scenery	1
I	prepare music	2
J	make costumes	4
K	dress rehearsals	2

Each activity needs a room except for activities G, I and J.

Choosing actors and dancers can be done in the same week. Rehearsals can begin after this.

Rehearsing the dancers cannot begin until the music has been prepared.

The scenery must be installed after rehearsals, but before dress rehearsals.

Making the costumes cannot start until after the actors and dancers have been chosen.

Everything must be ready for the dress rehearsals in the final two weeks of the 12-week preparation period.

- (i) Complete the table in your answer book by showing the immediate predecessors for each activity. [4]
- (ii) Draw an activity on arc network for these activities. [5]
- (iii) Mark on your network the early time and the late time for each event. Give the critical activities. [5]

It is discovered that there is a double booking and that one of the rooms will not be available after week 6.

- (iv) Using the space provided, produce a schedule showing how the pantomime can be ready in time for its first performance. [2]

Question 3:

- 5 A village amateur dramatic society is planning its annual pantomime. Three rooms in the village hall have been booked for one evening per week for 12 weeks. The following activities must take place. Their durations are shown.

Activity		Duration (weeks)
A	choose lead actors	1
B	choose rest of actors	1
C	choose dancers	1
D	rehearse lead actors	8
E	rehearse rest of actors	6
F	rehearse dancers	6
G	prepare scenery	6
H	install scenery	1
I	prepare music	2
J	make costumes	4
K	dress rehearsals	2

Each activity needs a room except for activities G, I and J.

Choosing actors and dancers can be done in the same week. Rehearsals can begin after this.

Rehearsing the dancers cannot begin until the music has been prepared.

The scenery must be installed after rehearsals, but before dress rehearsals.

Making the costumes cannot start until after the actors and dancers have been chosen.

Everything must be ready for the dress rehearsals in the final two weeks of the 12-week preparation period.

- (i) Complete the table in your answer book by showing the immediate predecessors for each activity. [4]
- (ii) Draw an activity on arc network for these activities. [5]
- (iii) Mark on your network the early time and the late time for each event. Give the critical activities. [5]
- It is discovered that there is a double booking and that one of the rooms will not be available after week 6.
- (iv) Using the space provided, produce a schedule showing how the pantomime can be ready in time for its first performance. [2]

- 4 The table lists tasks which are involved in adding a back door to a garage. The table also lists the duration and immediate predecessor(s) for each task. Each task is undertaken by one person.

Task		Duration (hours)	Immediate predecessor(s)
A	measure	0.5	–
B	manufacture frame and door	5	A
C	cut hole in wall	2	A
D	fit lintel and marble step	1.5	C
E	fit frame	1	B, C
F	fit door	1	E
G	repair plaster around door	1	E

- (i) Draw an activity on arc network for these activities. [5]
- (ii) Mark on your diagram the early time and the late time for each event. Give the minimum completion time and the critical activities. [6]
- (iii) Produce a schedule to show how two people can complete the project in the minimum time. [2]
- Soon after starting activity D, the marble step breaks. Getting a replacement step adds 4 hours to the duration of activity D.
- (iv) How does this delay affect the minimum completion time, the critical activities and the minimum time needed for two people to complete the project? [3]